Table 1: C3-regimes per system (sequence OOD). Change: paired relative change of the best KKL vs the best grey-box filter, ± 95% CI half-width (bold/underline: significant).

condition / system	KKL	GB	KKL vs GB	div. KKL/GB	forget KKL/GB (s)
clean					
battery	HyperKKL 0.0744	GB-EKF 0.0191	+295% ±190	0.00/0.00	–
bioreactor	HyperKKL 0.0753	GB-UKF 0.281	**-73%** ±13	0.00/0.00	–
duffing	HyperKKL 0.0536	GB-UKF 0.00775	+593% ±191	0.00/0.00	–
lorenz	HyperKKL 0.0308	GB-EKF-or 0.227	**-86%** ±8	0.00/0.00	–
quad_tank	HyperKKL 0.0717	GB-EKF 0.0142	+406% ±40	0.00/0.00	–
vdp	HyperKKL 0.0179	GB-EKF 0.0257	**-30%** ±15	0.00/0.00	–
poor_init					
battery	HyperKKL 0.0744	GB-EKF 0.0195	+288% ±186	0.00/0.00	3.31/2.8
bioreactor	HyperKKL 0.0753	GB-UKF 0.281	**-73%** ±13	0.00/0.00	4.51/0.68
duffing	HyperKKL 0.0536	GB-UKF 0.00775	+593% ±191	0.00/0.00	1.92/0.567
lorenz	HyperKKL 0.0308	GB-EKF-or 0.227	**-86%** ±8	0.00/0.00	1.02/0.681
quad_tank	HyperKKL 0.0717	GB-EKF 0.0142	+406% ±40	0.00/0.00	65.7/8.55
vdp	HyperKKL 0.0179	GB-EKF 0.0257	**-30%** ±15	0.00/0.00	2.19/0.556
noise10					
battery	HyperKKL 0.104	GB-EKF-or 0.0199	+421% ±90	0.00/0.00	–
bioreactor	HyperKKL 0.0753	GB-UKF 0.281	**-73%** ±13	0.00/0.00	–
duffing	HyperKKL 0.157	GB-EKF-or 0.0102	+1441% ±322	0.00/0.00	–
lorenz	HyperKKL 0.0467	GB-EKF-or 0.221	**-79%** ±5	0.00/0.00	–
quad_tank	HyperKKL 0.0994	GB-EKF-or 0.0156	+537% ±46	0.00/0.00	–
vdp	HyperKKL 0.0451	GB-EKF 0.0281	+61% ±32	0.00/0.00	–
outliers					
battery	HyperKKL 1.74	GB-EKF-or 0.0418	+4059% ±927	0.00/0.00	–
bioreactor	HyperKKL 0.136	GB-UKF 0.284	**-52%** ±39	0.00/0.00	–
duffing	HyperKKL 2.28	GB-EKF-or 0.0115	+19638% ±3963	0.00/0.00	–
lorenz	HyperKKL 0.572	GB-EKF-or 0.672	-14% ±27	0.00/0.01	–
quad_tank	HyperKKL 1.35	GB-EKF-or 0.0256	+5193% ±509	0.00/0.00	–
vdp	HyperKKL 0.697	GB-EKF-or 0.166	+319% ±19	0.00/0.00	–
plant_drift10					
battery	HyperKKL 0.0983	GB-EKF-or 0.0569	+73% ±35	0.00/0.00	–
bioreactor	HyperKKL 1.67	GB-UKF 1.85	**-9%** ±1	0.00/0.00	–
duffing	HyperKKL 0.0535	GB-EKF-or 0.0106	+405% ±120	0.00/0.00	–
lorenz	HyperKKL 0.171	GB-EKF-or 0.112	+53% ±10	0.00/0.00	–
quad_tank	HyperKKL 0.721	GB-EKF-or 1.56	**-53%** ±14	0.00/0.00	–
vdp	HyperKKL 0.0311	GB-EKF 0.0469	**-34%** ±5	0.00/0.00	–
burst_outage					
battery	HyperKKL 0.247	GB-EKF-or 0.0191	+1203% ±302	0.00/0.00	–
bioreactor	HyperKKL 0.0872	GB-UKF 0.284	**-69%** ±19	0.00/0.00	–
duffing	HyperKKL 0.51	GB-EKF-or 0.0084	+5973% ±543	0.00/0.00	–
lorenz	HyperKKL 0.36	GB-EKF-or 0.372	-3% ±5	0.00/0.00	–
quad_tank	HyperKKL 0.528	GB-EKF-or 0.0154	+3318% ±289	0.00/0.00	–
vdp	HyperKKL 0.318	GB-EKF-or 0.0431	+637% ±28	0.00/0.00	–